

Distributional Dynamic Programming in Economics

Mingli Chen* Yaolang Zhong†

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Abstract

This paper introduces a new family of decision-making models for agents in structural economic environments, in which agents evaluate the entire return distribution rather than only its expectation. We extend the traditional framework of expectation maximization by incorporating various measures over the distribution function, allowing for a more nuanced analysis of agent behavior under uncertainty. The proposed framework, termed *Distributional Dynamic Programming (DDP)*, generalizes conventional dynamic programming to focus on the *value distribution function*, from which optimal policies are derived. We develop the *Distributional Value Function Iteration (DVFI)* method, which iteratively applies the *distributional Bellman operator* and M -quantile projections to solve for the value distribution function. We apply this framework to two seminal models in economics: the McCall search model in labor economics and the Aiyagari model in macroeconomics. In the McCall model, mean-variance and quantile measures change the agent's value distribution and the implied decision to accept or reject wage offers; in the Aiyagari model, DVFI reproduces the value function iteration equilibrium when the mean measure is used, validating the method in a general-equilibrium setting.

1 Introduction

The economics literature has traditionally viewed agents as expectation maximizers in dynamic decision-making (see, for example, [Samuelson, 1983](#); [Koopmans, 1960](#); [Brock and Mirman, 1972](#)). This approach offers a straightforward method to compare different options and outcomes. However, substantial evidence indicates that human decision-making is more nuanced, particularly under uncertainty ([de Castro et al., 2022](#)). Assuming that individuals care only about the first moment of the return distribution may not fully capture the complexity of their behavior.

*Department of Economics, The University of Warwick

†Graduate School of Economics, The University of Tokyo

Consider a scenario where you need to catch an early-morning flight for a conference and want the shortest trip to the airport to maximize your sleep. Route A consistently takes 1.5 hours, while Route B usually takes 1.2 hours but has a 10% chance of being closed for maintenance, potentially extending the trip to over 3 hours and causing you to miss the flight. An expectation maximizer would likely choose Route B for its shorter average time, but someone who places significant weight on the risk of missing the flight might prefer Route A.

The traditional approach to incorporating such risk aversion involves adopting a utility function over travel time, with a risk-aversion parameter that adjusts the curvature of the function. In this way, the expected utility, rather than the travel time alone, reflects the agent’s risk preferences. However, defining such a utility function for a particular problem is not always feasible or straightforward.

This paper proposes a family of alternative decision-making models for agents in structural environments. Rather than adopting a utility function, we directly model the distribution of returns and encode the agent’s preferences through a measure over this distribution—for instance, one that balances the trade-off between expectation and variance. In our framework, agents consider not the expected return but the entire return distribution, and each decision model is characterized by the maximization of a specific measure of the distribution function. This approach encompasses the traditional expectation-maximization framework as a special case, when the measure corresponds to the first moment, and thereby introduces a new dimension to the analysis.

More specifically, this paper considers the Markov decision process faced by an agent, as is common in many dynamic economic models. We introduce *Distributional Dynamic Programming (DDP)* by generalizing the discussion in conventional dynamic programming, which typically focuses on the value function, to consider the so-called *value distribution function* of the agent. Analogous to the case where an agent derives the optimal policy from the value function, the optimal policy is now derived from the value distribution function, with respect to a pre-chosen measure that maps the value distribution function to a scalar value. We propose various measures that reflect different preferences over the distribution.

We then discuss the properties of the value distribution function, showing that it is the fixed point of the *distributional Bellman operator*, just as the value function is the fixed point of the conventional Bellman operator. We consider the p -Wasserstein distance as a metric in the function space, noting that the distributional Bellman operator has been proven to be a contraction mapping with modulus equal to the discount factor. Hence, we can obtain the fixed point by iteratively applying the operator to an initial guess.

However, for implementation, we must address the practical issue of how to characterize a distribution. In this paper, we consider the M -quantile parameterization, which characterizes a distribution by M equally weighted particles. We then introduce a projection method that ensures that the projected distribution has the smallest 1-Wasserstein distance to the original

distribution, thus achieving a w_1 -projection.

Finally, we combine all these components and propose an algorithm to implement the distributional Bellman operator, alternating with the M -quantile projection, until the value distribution function converges and the policy function is obtained. We call this approach *Distributional Value Function Iteration (DVFI)*, by analogy with Value Function Iteration (VFI) in the context of handling the value function.

For applications, we consider two foundational models in the economics literature: the McCall search model (McCall, 1970) in labor economics and the Aiyagari model (Aiyagari, 1994) in macroeconomics. In the former, we extend the agent’s decision-making process from merely trading off between accepting and rejecting a randomly drawn offer based on expected cumulative discounted income to scenarios where the agent is risk-averse. This includes cases where the agent imposes varying degrees of penalty on income variance if the current offer is rejected, as well as situations where the agent considers specific quantile levels of the return. In the latter application, where general equilibrium is characterized by the wealth distribution among agents, we examine whether DVFI produces the same solution as VFI when the first moment is chosen as the measure of agents’ preferences over the distribution. In future work, we will explore how different preferences, determined by the choice of measure in the value distribution function, impact the stationary distribution of the economy. This will be compared to the benchmark case where the agent is an expectation maximizer.

The remainder of the paper is structured as follows: Section 2 provides the literature review; Section 3 introduces the technical aspects of the Distributional Dynamic Programming approach; Section 4 presents the first computational example, the McCall search model, and Section 5 discusses the second example, the Aiyagari model. In Section 6, we briefly outline various directions for future work, and finally, Section 7 concludes the paper.

2 Literature Review

This paper is closely related to the work of de Castro and Galvao (2019), which developed a dynamic model where agents maximize the stream of future τ -quantile utilities. In subsequent work, de Castro et al. (2023) studied numerical methods for the quantile preference model, and de Castro et al. (2022) provided empirical evidence supporting agents’ quantile maximization behavior. Our work generalizes these results by allowing agents to maximize a particular measure of the distribution of returns. The two approaches coincide when the chosen measure is a quantile measure.

This paper builds on ideas from a rapidly growing body of literature on distributional approaches in reinforcement learning (Bellemare et al., 2017; Dabney et al., 2018b,a; Lyle et al., 2019). Notably, much of the technical material in Section 3 is drawn from the recent

survey by [Bellemare et al. \(2023\)](#). The main contributions of this paper are twofold. First, we apply distributional approaches within economic environments, which feature multiple sources of uncertainty. The discussion of risk-averse agents in economics dates back several decades, highlighting the necessity of distributional decision-making models in this context. Second, we explore distributional decision-making within the dynamic programming framework, which is prevalent in economics, where agents are typically assumed to have knowledge of the state transition dynamics. This contrasts with the deep reinforcement learning literature mentioned above, where agents often learn the environment’s transitions through trial and error, either in real-world experiments or simulators.

3 Methodology

3.1 From Value Function to Value Distribution Function

To begin with, we focus on the discrete-time Markov decision process (MDP), which is a tuple $(\mathcal{S}, \mathcal{A}, \gamma, P_{\mathcal{S}}, P_{\mathcal{R}})$, where $\mathcal{S}$ is the state space, with state $S_t \in \mathcal{S}$; $\mathcal{A}$ is the action space, with action $A_t \in \mathcal{A}$; γ is the discount factor; and $P_{\mathcal{S}}$ and $P_{\mathcal{R}}$ are the transition kernel and reward distribution functions, respectively. Specifically, $R_t \sim P_{\mathcal{R}}(\cdot \mid S_t, A_t)$ and $S_{t+1} \sim P_{\mathcal{S}}(\cdot \mid S_t, A_t)$, where we assume that the reward R_t and the next state S_{t+1} are independent given S_t and A_t . Since the reward and transition are conditional on the action, the description of the MDP is complete given a specific policy, which is a mapping $\pi : \mathcal{S} \rightarrow \mathbb{P}(\mathcal{A})$ from states to probability distributions over actions such that $A_t \sim \pi(\cdot \mid S_t)$.

We are interested in the accumulated discounted return of an agent obtained by following a particular policy π from a given state s in an MDP:

$$G^\pi(s) = \sum_{t=0}^{\infty} \gamma^t R_t, \quad S_0 = s.$$

Note that since $\{R_t\}_{t=0}^{\infty}$ is a sequence of random variables, their sum G^π is itself a random variable. If rewards are bounded, $R_t \in [R_{\min}, R_{\max}]$, then $G^\pi \in [\frac{R_{\min}}{1-\gamma}, \frac{R_{\max}}{1-\gamma}]$. A typical object of interest is the expectation of the random variable G^π and, accordingly, the optimal policy π^* that maximizes it, which gives the familiar state value function:

$$V^*(s) = \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s \right] = \mathbb{E}[G^*(s)]. \quad (1)$$

We use the superscript $*$ to denote any function or variable associated with the optimal policy π^* . Another function of particular interest, especially in the computer science literature, is the state-action value function, commonly known as the “ Q -function” ([Watkins and](#)

Dayan, 1992; see also Sutton, 2018):

$$Q^*(s, a) = \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s, A_0 = a \right] = \mathbb{E}[G^*(s, a)], \quad (2)$$

where, with a slight abuse of notation, we use G to now denote $G^*(s, a)$, the accumulated discounted return obtained by following the optimal policy π^* from a given state s and action a , and the Q -function represents its expectation. The Q -function has some computational convenience since we can easily recover V^* and obtain the optimal policy function π^* from Q^* :

$$V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a) \quad (3a)$$

$$\pi^*(s) = \arg \max_{a \in \mathcal{A}} Q^*(s, a) \quad (3b)$$

Therefore, from this point forward, we will focus our discussion on the Q -function. Whenever we refer to the “value function,” we specifically refer to the state-action value function, the Q -function. However, note that all the results presented have an analogous version for the standard value function V .

While much of the literature focuses on the expectation of random returns, this paper broadens the discussion to include any measure of the random variable G^* , or equivalently, its distribution. Specifically, we introduce the notation η^* , known as the *value distribution function*, where $\eta^*(s, a)$ represents the distribution of $G^*(s, a)$ for a given state-action pair (s, a) .

Next, we introduce the measure function $\rho : \mathcal{P}(\mathbb{R}) \rightarrow [-\infty, \infty)$, which maps a distribution function to a scalar. This measure function characterizes the agent’s specific preference over distributions, allowing us to compare and rank different distributions based on the chosen measure.

Our objective is to seek an optimal policy π^* such that for each state s we maximize a generalized value of the return, measured by a particular ρ :

$$\pi^*(s) = \arg \max_a \rho(\eta^*(s, a)). \quad (4)$$

That is, we derive the optimal policy function π^* once we have the value distribution function η^* . In the next subsection we describe how to compute η^* . Below, we consider some particular families of measures:

- Measures over moments of the distribution: let the moments of a distribution up to order L be denoted as $m = (m_1, \dots, m_L)$. A measure $\rho_{\text{moment}} = h(m)$ can take a

specific functional form that reflects the agent’s trade-off between different moments of interest. For example, the mean, $\rho_M(\eta) = \mathbb{E}(\eta) = m_1$, corresponds to the standard value function; the mean–variance trade-off, as commonly discussed in the finance literature, penalizes dispersion: letting $\lambda > 0$ be the penalty parameter on variance, $\rho_{MV}^\lambda(\eta) = \mathbb{E}(\eta) - \lambda\sqrt{\mathbb{V}(\eta)} = m_1 - \lambda\sqrt{m_2}$.

- Measures over specific quantile levels/Value-at-Risk: let $\tau > 0$. The measure ρ_Q^τ represents the τ -quantile of the distribution, defined as $\rho_Q^\tau(\eta) = F_\eta^{-1}(\tau)$.
- Measures based on entropy: let f_η denote the PDF of η . The entropy measure ρ_H is defined as $\rho_H(\eta) = -\int f_\eta(x) \log f_\eta(x) dx$. This measure captures the degree of uncertainty or dispersion in the distribution.
- Measures over the distance from a benchmark: let $\hat{\eta}$ be a benchmark distribution that the agent prefers η to closely resemble, according to a metric in function space $g : \mathcal{P}(\mathbb{R}) \times \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$. The measure $\rho_{\hat{\eta}}^g$ is then defined as $\rho_{\hat{\eta}}^g(\eta) = -g(\eta, \hat{\eta})$, so that maximizing the measure selects distributions close to the benchmark.

3.2 Distributional Bellman Operator

To write down the recursive form of an MDP problem, we drop the time subscript t and use X' to denote the variable X in the next period. It is well known that, under mild conditions, Q^* is the unique fixed point of the celebrated Bellman operator (Bellman, 1966):

$$\begin{aligned} (\mathcal{T}Q)(s, a) &= r(s, a) + \gamma \int_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} Q(s', a') f(s' | s, a) ds', \quad s \in \mathcal{S}, a \in \mathcal{A} \\ Q^* &= \mathcal{T}Q^* \end{aligned} \tag{5}$$

Given the current estimated Q -value of a particular (s, a) pair, the Bellman operator consists of three operations to obtain the new estimate:

- Maximization: conditional on s' , we choose the a' that gives us the maximum value.
- Integration: we integrate over s' governed by the transition function with (s, a) .
- Scaling: we shift the integrated value by defining the bootstrap function $b_{r, \gamma} : \mathbb{R} \rightarrow \mathbb{R}$, indexed by the reward $r \in \mathbb{R}$ and discount factor $\gamma \in [0, 1)$, such that $b_{r, \gamma}(z) = r + \gamma z$.

Intuitively, being the fixed point of the Bellman operator is the constraint imposed on Q^* , the first moment of the random variable G^* . To generalize this idea, an extended version of the Bellman operator, the so-called *distributional Bellman operator* (Bellemare et al., 2017), has been proposed to characterize η^* , the whole distribution of G^* :

$$(\mathcal{T}\eta)(s, a) = \mathcal{B}_{r,\gamma} \int_{s' \in \mathcal{S}} \eta(s', a') f(s' \mid s, a) ds' \quad (6a)$$

$$\text{where } a' = \arg \max_{a' \in \mathcal{A}} \rho(\eta(s', a')) \quad (6b)$$

$$\eta^* = \mathcal{T}\eta^* \quad (6c)$$

where we use the same symbol $\mathcal{T}$ for the operator for simplicity of notation. Analogous to the case of the benchmark Bellman operator, we can understand the distributional Bellman operator as a composition of three operations:

- Maximization over ρ : conditional on s' , we choose the a' that gives us the maximum value of the function under the measure ρ .
- Integration: the integral over next states becomes a mixture of distributions η , which is technically more challenging; we discuss this in more detail in the next subsection.
- Scaling: similarly to the case above, but now $\mathcal{B}_{r,\gamma}$ is a bootstrap operator on the integrated distribution, which means that we apply the mapping $b_{r,\gamma} : z \rightarrow r + \gamma z$ to every realization of η .

We say that η^* satisfies *the distributional Bellman equation* if it is the solution to (6), analogous to how Q^* is the solution to the Bellman equation. However, this does not provide a method for obtaining such a solution. In the case of Q^* , we know that it can be found by iteratively applying the Bellman operator to an initial guess Q until convergence, since the Bellman operator is a contraction mapping under the supremum norm. To generalize this result and solve for η^* by iteratively applying the distributional Bellman operator to an initial guess η , two conditions must be met: (1) a defined metric space for the value distribution function η and (2) proof that the distributional Bellman operator is a contraction mapping in this function space. In the following, we briefly summarize results from the current literature, focusing on one of the metrics commonly used in the space of distribution functions, the p -Wasserstein distance.

Definition 1 (Kantorovich, 1960). Let $\eta \in \mathcal{P}(\mathbb{R})$ be a distribution function with its CDF denoted by F_η ; its generalized inverse is given by $F_\eta^{-1}(\tau) = \inf_{z \in \mathbb{R}} \{z : F_\eta(z) \geq \tau\}$. Let $p \in [1, \infty)$. The p -Wasserstein (w_p) distance is a function $g_p : \mathcal{P}(\mathbb{R}) \times \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ given by

$$g_p(\eta, \eta') = \left(\int_0^1 |F_\eta^{-1}(\tau) - F_{\eta'}^{-1}(\tau)|^p d\tau \right)^{1/p}.$$

Graphically, the Wasserstein distances between two probability distributions measure the area between their CDFs, with values along the abscissa taken to the p -th power. In terms

of the value distribution function η , we consider the supremum p -Wasserstein distance over the (s, a) pairs:

Definition 2. Let $p \in [1, \infty]$. The supremum p -Wasserstein distance $\bar{g}_p$ between two value distribution functions $\eta, \eta' \in \mathcal{P}(\mathbb{R})^{S \times A}$ is defined by

$$\bar{g}_p(\eta, \eta') = \sup_{s \in S, a \in A} g_p(\eta(s, a), \eta'(s, a)).$$

For a fixed policy π , define the policy-evaluation operator $\mathcal{T}^\pi$ analogously to (6), except that the next-period action is drawn from the given policy, $a' \sim \pi(\cdot \mid s')$, instead of being selected greedily with respect to ρ . Based on the supremum p -Wasserstein distance, it has been shown that this operator is contractive, as follows:

Proposition 1. The distributional Bellman operator $\mathcal{T}^\pi$ is a contraction mapping on $\mathcal{P}(\mathbb{R})^{S \times A}$ in the supremum p -Wasserstein distance for all $p \in [1, \infty]$. More precisely,

$$\bar{g}_p(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \eta') \leq \gamma \bar{g}_p(\eta, \eta'),$$

for all $\eta, \eta' \in \mathcal{P}(\mathbb{R})^{S \times A}$.

The proof can be found in Chapter 4 of [Bellemare et al. \(2023\)](#). Note that the contraction result is stated for the policy-evaluation operator $\mathcal{T}^\pi$; in the algorithm below we apply the control operator $\mathcal{T}$ of (6), which replaces the fixed policy with the greedy selection under ρ . This proposition demonstrates that iteratively applying the distributional Bellman operator to an initial guess of the value distribution function η_0 guarantees convergence. However, two practical challenges remain:

- Characterizing a distribution function on a computer.
- Ensuring that the chosen characterization is closed under the distributional Bellman operator, which requires applying a projection to ensure that the new estimate belongs to the same family of characterizations.

These issues are addressed in the next subsection.

3.3 Parameterization and Projection

Denote by $\mathcal{F}$ a particular parametric family. To approximate an arbitrary distribution $\eta \in \mathcal{P}(\mathbb{R})$ by $\mathcal{F}$, we introduce the projection operator $\mathcal{P} : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{F}$. Intuitively, $\mathcal{P}$ chooses a distribution $\hat{\eta}$ from $\mathcal{F}$ to minimize the difference from η measured by a given metric d :

$$\mathcal{P}(\eta) = \arg \min_{\hat{\eta} \in \mathcal{F}} d(\eta, \hat{\eta}). \quad (7)$$

In this case, we call $\mathcal{P}$ a d -projection. Here, we consider minimizing the 1-Wasserstein (w_1) distance and one of the parameterizations of the distribution function that is suited to this metric, the M -quantile parameterization.

Definition 3. *The M -quantile parameterization of a distribution function η is given by the set:*

$$\mathcal{P}_Q^M(\eta) \in \mathcal{F}_Q^M = \left\{ \frac{1}{M} \sum_{m=1}^M \delta_{\theta_m} : \theta_m \in \mathbb{R} \right\} \quad (8)$$

That is, the M parameters $\{\theta_m\}_{m=1}^M$ characterize the locations of M equally weighted particles. The following proposition gives the optimal selection of θ when we choose the 1-Wasserstein distance:

Proposition 2. *Let $\eta \in \mathcal{P}(\mathbb{R})$. The M -quantile probability distribution whose parameters are given by*

$$\theta_m = F_\eta^{-1} \left(\frac{2m-1}{2M} \right), \quad m = 1, \dots, M \quad (9)$$

is a w_1 -projection of η onto $\mathcal{F}_Q^M$. We denote by $\tau_M = \{\frac{2m-1}{2M}\}_{m=1}^M$ the quantile levels determined by the choice of M .

The proposition states that each interval $[\frac{m-1}{M}, \frac{m}{M}]$, $m = 1, \dots, M$, is characterized by its midpoint $\frac{2m-1}{2M}$ quantile. Figure 1 illustrates the case where we approximate a random distribution η by the step function $\hat{\eta}$, choosing $M = 5$.

The M -quantile parameterization requires knowledge of the exact values associated with τ_M . On the other hand, there is an integration step in the distributional Bellman operator over the integrands indexed by the next state s' and weighted by the transition function $f(s'|s, a)$. By applying the distributional Bellman operator in each iteration, even if the integrands are M -quantile projected (which can be achieved by initialization), the integral is not guaranteed to be M -quantile projected. In other words, the M -quantile projection is not guaranteed to be closed under the distributional Bellman operator.

We use an example to demonstrate the implementation of the integration and subsequent projection of a set of M -quantile projected integrands. Again, we choose $M = 5$ so that $\tau_M = (0.1, 0.3, 0.5, 0.7, 0.9)$. There are two distributions $\eta_1, \eta_2 \in \mathcal{F}_Q^M$ such that η_1 is characterized by $X_1 = (1, 3, 5, 7, 9)$ and η_2 is characterized by $X_2 = (0, 2, 6, 10, 10)$. η_3 is a mixture of η_1 and η_2 with weights $W = (0.4, 0.6)$: $\eta_3 = W_1\eta_1 + W_2\eta_2$. Our goal is the M -quantile projection of η_3 .

Since each function is characterized by equally weighted particles (in this case, $\frac{1}{M} = 0.2$ for each), the mixture distribution has each particle with a new weight equal to the original

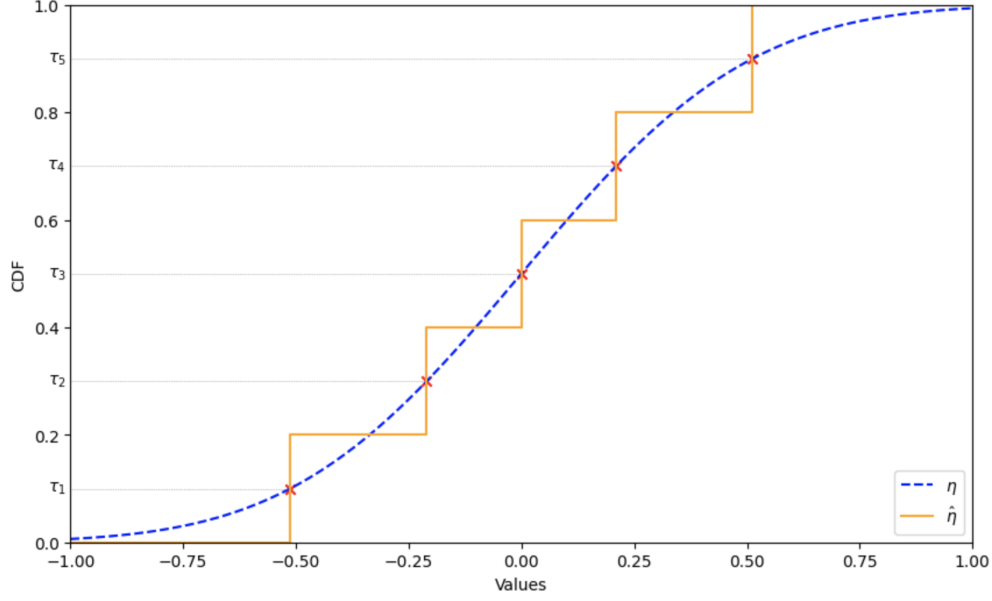


Figure 1: An example of quantile projection with $M = 5$. The blue dashed curve is the CDF of the original distribution η ; the orange solid step function is the CDF of the projected distribution $\hat{\eta}$. The unit interval is divided into the intervals $([0, 0.2], [0.2, 0.4], [0.4, 0.6], [0.6, 0.8], [0.8, 1.0])$, and the particles (red crosses) are placed at the associated midpoint quantiles $(\tau_1, \dots, \tau_5) = (0.1, 0.3, 0.5, 0.7, 0.9)$.

weight times the weight in W . Then, by ordering the particles according to their values, we can find the corresponding value for each quantile level in τ_M . Specifically, we first sort the combined set of values $X_3 = (X_1, X_2)$, then order the associated weights for each particle: for example, the mass of each particle in η_2 is $W_2/M = 0.12$; and finally we search for the value corresponding to each quantile level in τ_M in the new distribution. Figure 2 gives an illustration of this example.

We generalize the implementation for the case of K value distribution functions $\{\eta_k\}_{k=1}^K \in \mathcal{F}_Q^M$, where $K = |\mathcal{S}|$ is the size of the state space, as stated in Algorithm 1 below.

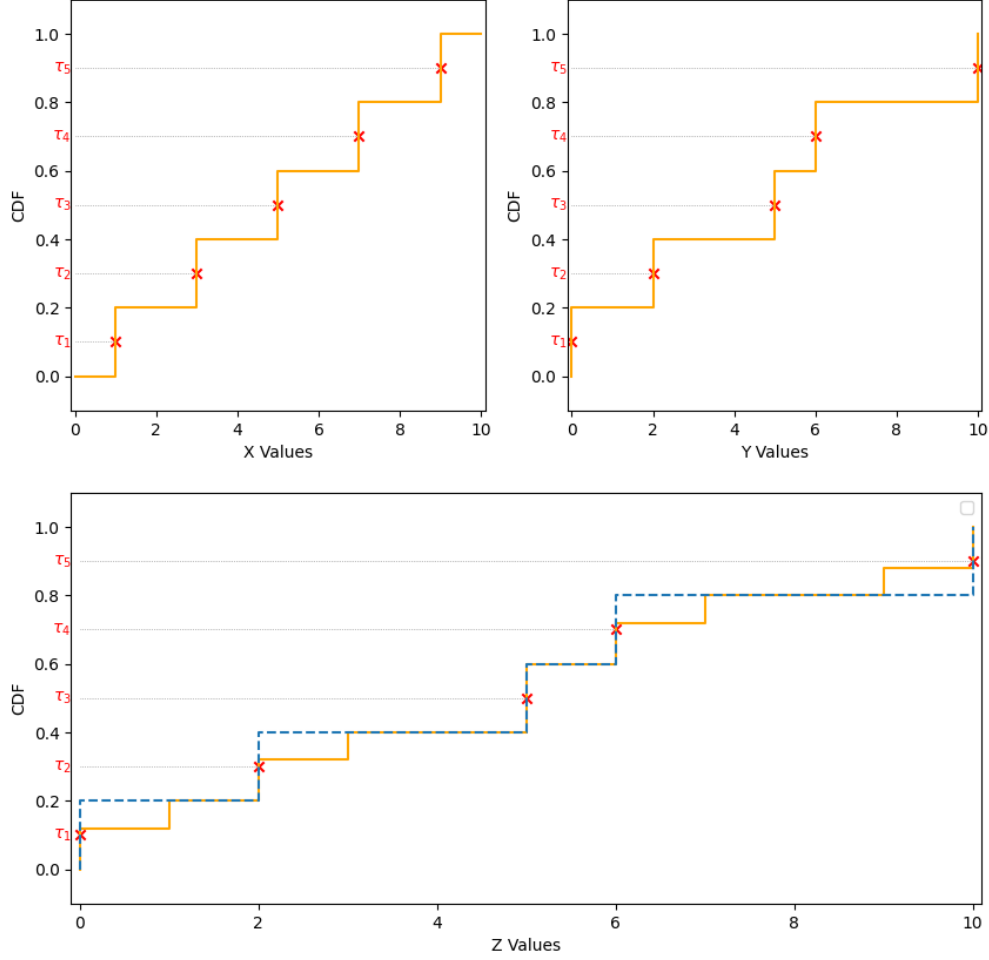


Figure 2: Illustration of the integration and subsequent projection of M -quantile projected integrands. $M = 5$ and $\tau_M = (0.1, 0.3, 0.5, 0.7, 0.9)$. Upper left: η_1 with values $X_1 = (1, 3, 5, 7, 9)$. Upper right: η_2 with values $X_2 = (0, 2, 6, 10, 10)$. Bottom: blue solid line: the mixture η_3 before projection; orange dashed line: the mixture η_3 after projection.

Algorithm 1 Integration and Projection of M -Quantile Projected Integrands

- 1: **Parameters:** K : number of integrand functions, M : number of quantile levels to characterize the distributions
 - 2: **Input:** M -quantile levels $\tau_M = \{\frac{2m-1}{2M}\}_{m=1}^M$, Distributions $\{\eta_k\}_{k=1}^K \in \mathcal{F}_Q^M$ characterized by $\{X_k = (x_{mk})_{m=1}^M\}_{k=1}^K$, Weights $W = \{W_k\}_{k=1}^K$
 - 3: **Output:** M -quantile projection of the mixture distribution $\mathcal{P}_Q^M(\eta_{\text{mix}})$, where $\eta_{\text{mix}} = \sum_{k=1}^K W_k \eta_k$
 - 4: Concatenate values $X_{\text{mix}} = (X_1, X_2, \dots, X_K)$
 - 5: Compute new weights for each particle in X_{mix} as the original weight times the weight in W
 - 6: Sort X_{mix} in ascending order as $\bar{X}_{\text{mix}}$
 - 7: Order the associated weights for each particle in $\bar{X}_{\text{mix}}$
 - 8: Find in $\bar{X}_{\text{mix}}$ the value corresponding to each quantile level for a new vector $\tilde{X}_{\text{mix}}$
 - 9: **Return:** $\tilde{X}_{\text{mix}}$
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3.4 Implementation

In this subsection, we collect all the discussions so far to describe the workflow for the distributional approach, specifically choosing a particular measure ρ and the M -quantile characterization and projection for the distribution function. Again, the goal is to solve an MDP, and after initializing a value distribution function $\eta : \eta(s, a) \in \mathcal{F}_Q^M, \forall s \in \mathcal{S}, a \in \mathcal{A}$, we iteratively apply the distributional Bellman operator as in Equation (6), together with the M -quantile projection:

$$(\mathcal{T}\eta)(s, a) = \mathcal{B}_{r, \gamma} \mathcal{P}_Q^M \left(\int_{s' \in \mathcal{S}} \eta(s', a') f(s' | s, a) ds' \right) \quad (10a)$$

$$\text{where } a' = \arg \max_{a' \in \mathcal{A}} \rho(\eta(s', a')) \quad (10b)$$

$$\eta^* = \mathcal{T}\eta^* \quad (10c)$$

Algorithm 2 states the full procedure.

Algorithm 2 Distributional Dynamic Programming with M -Quantile Projection

- 1: **Parameters:** An MDP tuple $(\mathcal{S}, \mathcal{A}, \gamma, P_S, P_R)$; number of quantile levels M and the associated $\tau_M = \left\{ \frac{2m-1}{2M} \right\}_{m=1}^M$; measure ρ ; a metric d and threshold ϵ for convergence checking; maximum number of iterations *MaxIter*
 - 2: **Initialization:** The M -quantile projected value distribution function $\eta^0 \in \mathcal{F}_Q^M : \mathcal{S} \times \mathcal{A} \times \tau_M \rightarrow \mathbb{R}$ stored in a $|\mathcal{S}| \times |\mathcal{A}| \times M$ matrix
 - 3: **for** $i = 0$ to *MaxIter* **do**
 - 4: **Step 1: Maximization over ρ**
 - 5: **for** each $s' \in \mathcal{S}$ **do**
 - 6: Choose $\eta^i(s', a')$ such that $a' = \arg \max_{a' \in \mathcal{A}} \rho(\eta^i(s', a'))$
 - 7: **end for**
 - 8: **for** each $(s, a) \in \mathcal{S} \times \mathcal{A}$ **do**
 - 9: **Step 2: Integration and Projection**
 - 10: Implement Algorithm 1 with distributions $\{\eta^i(s', a')\}_{s' \in \mathcal{S}}$ and weights $\{f(s' | s, a)\}_{s' \in \mathcal{S}}$
 - 11: **Step 3: Scaling**
 - 12: Apply the bootstrap operator $\mathcal{B}_{r, \gamma}$ to the outcome from Step 2 to obtain $\eta^{i+1}(s, a)$ for the next iteration
 - 13: **end for**
 - 14: **Check:** If $d(\eta^i, \eta^{i+1}) < \epsilon$, set $\eta^* = \eta^{i+1}$, break, and report convergence
 - 15: **end for**
 - 16: If convergence, return η^* and the associated policy function $a^*(s) = \arg \max_{a \in \mathcal{A}} \rho(\eta^*(s, a))$
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In the next two sections, we apply the distributional methods to two classical structural models. Specifically, in Section 4, we examine the McCall search model in labor economics, exploring how an agent’s choice of distribution measure ρ influences their value distribution function and, consequently, their reservation wage when deciding whether to accept an offer. In Section 5, we turn to the Aiyagari model in macroeconomics to verify that the distributional approach reproduces the benchmark general-equilibrium solution when the mean measure is used, a prerequisite for studying how the choice of ρ affects the wealth distribution. For both models, we use the calibration from Sargent and Stachurski (2014), which offers a simplified version that highlights the key features of the models and provides source code for a fast benchmark VFI solution.

4 Case 1: McCall Search Model

The McCall search model (McCall, 1970) is a fundamental framework in labor economics that analyzes the decision-making process of unemployed workers as they search for job offers. This model addresses the trade-off between accepting a current job offer and continuing to search for potentially better opportunities in the future. It is particularly useful for understanding the dynamics of unemployment, job search behavior, and the impact of unemployment benefits on the job market.

4.1 Model Setup and Benchmark Solution

In each period, an unemployed worker draws a wage offer w_t from $F(\cdot)$ and maximizes the expected discounted sum of lifetime income, $\mathbb{E} \sum_{t=0}^{\infty} \gamma^t y_t$, where income y_t equals the wage w_t when employed and the constant unemployment compensation c when unemployed.

In recursive form, the state space is $\mathcal{S} = [w_{\min}, w_{\max}]$ and the action space is $\mathcal{A} = \{A, R\}$, indicating acceptance or rejection. The Q -function satisfies the following system of equations:

$$Q(w, A) = \frac{w}{1 - \gamma}, \quad (11a)$$

$$Q(w, R) = c + \gamma \int_{w' \in \mathcal{S}} \max_{a' \in \{A, R\}} Q(w', a') dF(w'), \quad (11b)$$

where $\frac{w}{1-\gamma}$ represents the present value of accepting the wage offer w , and the right-hand side of Equation (11b) represents the value for which the agent receives the unemployment compensation c and in the next period compares acceptance and rejection for each potential offer w' .

For implementation, we take two steps. First, we discretize the state space into J grid

points, $\{w_1, w_2, \dots, w_J\}$, indexed by j , with the corresponding probability mass $\{\mathbb{P}(w_1), \dots, \mathbb{P}(w_J)\}$. Second, we make use of the fact that w_t is drawn i.i.d. each period, so $Q(w, R)$ is constant across w ; we denote it by

$$Q_R := Q(w, R) \quad \forall w \in \mathcal{S}.$$

Since $Q(w, A)$ is already determined by Equation (11a), solving the system of equations (11) reduces to the following iteration:

$$Q_R^{i+1} = c + \gamma \sum_{j=1}^J \max\left\{\frac{w_j}{1-\gamma}, Q_R^i\right\} \mathbb{P}(w_j). \quad (12)$$

4.2 The Distributional Approach

Next, we introduce the distributional approach for this problem. Our target is the value distribution function η that satisfies the following system of equations:

$$\eta(w, A) = \delta_{\frac{w}{1-\gamma}} \quad (13a)$$

$$\eta(w, R) = \mathcal{B}_{c,\gamma} \int_{w' \in \mathcal{S}} \eta(w', a') dF(w') \quad (13b)$$

$$\text{where } a' = \arg \max_{a' \in \{A, R\}} \rho(\eta(w', a')) \quad (13c)$$

which states that once the agent chooses to accept the offer w , the return is determined and hence distributed according to the Dirac measure $\delta_{\frac{w}{1-\gamma}}$. On the other hand, the distribution of rejecting the offer w is a weighting of a continuum of distributions indexed by the next offer w' , conditional on which the action a' is chosen according to a selection criterion characterized by the measure ρ between two distributions.

Again we discretize the state space and make use of the fact that $\eta(w, R)$ is invariant across w ; we denote it by

$$\eta_R := \eta(w, R) \quad \forall w \in \mathcal{S}.$$

To apply Algorithm 2 we consider the M -quantile projection so that η_R is characterized by M particles associated with the quantile levels τ_M . On the other hand, $\eta(w, A)$ is represented by an M -vector with all elements $\frac{w}{1-\gamma}$. As in the case of the Q -function, the system of equations (13) reduces to the following iteration:

$$\eta_R^{i+1} = \mathcal{B}_{c,\gamma} \mathcal{P}_Q^M \sum_{w' \in \mathcal{S}} \kappa_\rho(\delta_{\frac{w'}{1-\gamma}}, \eta_R^i) \mathbb{P}(w'), \quad (14)$$

where $\kappa_\rho(\eta_1, \eta_2)$ returns whichever of the two distributions has the higher value of the measure ρ .

4.3 Calibration and Results

Table 1 lists the structural parameters and hyperparameters chosen for the benchmark case.

Parameter	Definition	Value
γ	Discount factor	0.99
c	Unemployment compensation	25
$w_{\min}$	Lower bound of state space	10
$w_{\max}$	Upper bound of state space	60
J	Number of grid points in discretization	11
F	Distribution of wage w	Beta-Binomial($n = 10, \alpha = 200, \beta = 100$)

Table 1: Parameters of the McCall search model.

Figure 3 illustrates the wage distribution. The distribution is asymmetric and left-skewed, with little mass at low wages. The Beta-Binomial support $\{0, 1, \dots, n\}$ maps onto the $J = n + 1 = 11$ grid points spanning $[w_{\min}, w_{\max}]$.

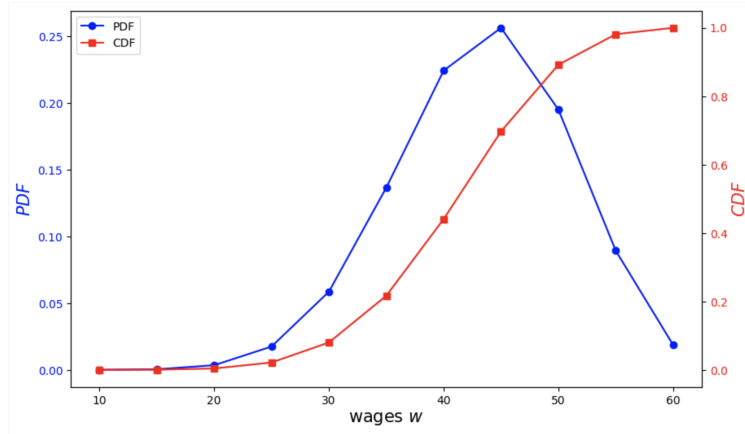


Figure 3: PDF and CDF of the wage distribution, Beta-Binomial($n = 10, \alpha = 200, \beta = 100$), over the $J = 11$ grid points spanning $[w_{\min}, w_{\max}] = [10, 60]$.

Figure 4a shows the benchmark result where we choose $M = 100$. Specifically, the blue line represents η_R in its CDF form. Note that while the theoretical lower bound of the

support of η_R is $\frac{w_{\min}}{1-\gamma} = 1000$, determined by accepting the lowest wage offer, Algorithm 2 returns an actual lower bound of $\frac{c}{1-\gamma} = 2500$: under the optimal policy, the agent never accepts offers worth less than the option of remaining unemployed indefinitely, so the return distribution is truncated below at this value. The computed support therefore matches the theoretical prediction. On the other hand, Algorithm 2 returns the same upper bound as the theoretical one, $\frac{w_{\max}}{1-\gamma} = 6000$, which is achievable by accepting the highest wage offer.

The CDF of η_R is much steeper than the wage CDF in Figure 3: the return distribution is more concentrated than the wage distribution, because the option to reject compresses the lower tail. For instance, the mean wage is approximately 43.3, while the mean of η_R , interpreted as the expected return if the offer is rejected, is 5228.43, substantially higher than $\frac{43.3}{1-\gamma} = 4330$. In Figure 3, the 20% quantile of the wage offer is around 40, whereas the 20% quantile of η_R is over 5000. This indicates that the mapping from the wage distribution (state variable) to the value function is not affine, which is attributable to the option of rejection in the structural model.

We compare the mean of η_R with Q_R obtained from the value function iteration in Section 4.1, which are 5228.43 and 5322.27, respectively. The difference is due to the approximation error introduced by the projection in each iteration, but is relatively small compared to the magnitude of the values, with a relative difference of only $\frac{5322.27 - 5228.43}{5228.43} = 1.7\%$. Figure 4 demonstrates that the approximation error in the value has no significant impact on the derived policy function: both approaches yield the same action for each wage offer.

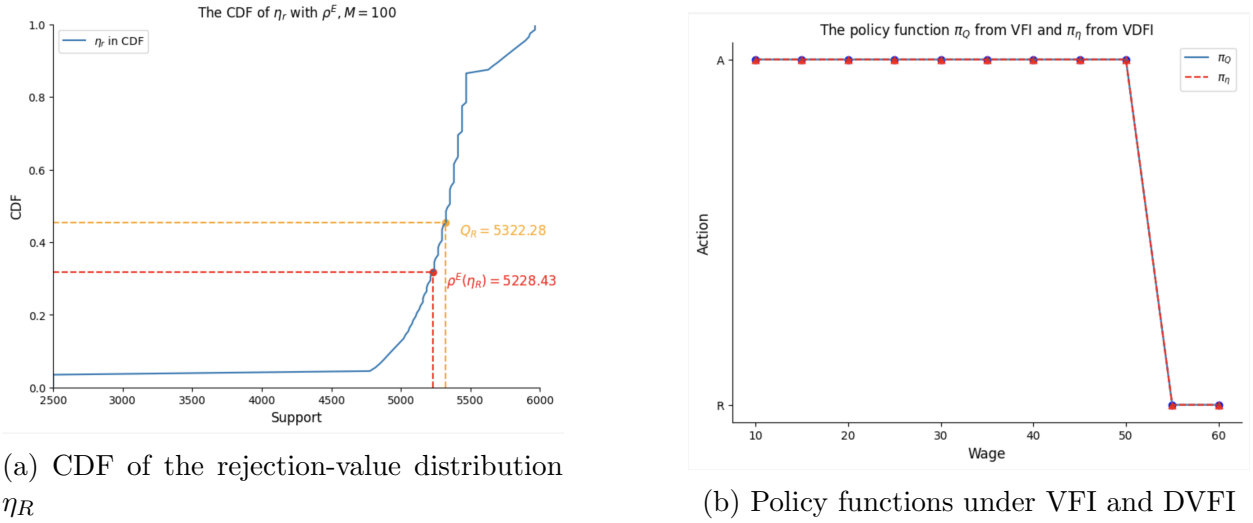


Figure 4: Benchmark results for the mean measure with $M = 100$. Panel (a) shows the CDF of the rejection-value distribution η_R . Panel (b) shows the policy functions derived from the Q -function (VFI) and from η_R (DVFI); both prescribe the same acceptance decision for each wage offer.

As a robustness check, Figure 5 demonstrates the relative differences between $\rho(\eta_R)$ and Q_R for a range of options $M \in \{10, 20, 50, 100, 200, 1000\}$. The results suggest that the M -

quantile projection may cause an underestimation of the mean of the distribution compared to the VFI result. The underlying reasons for this discrepancy require further investigation, which will be addressed in future extensions of this work. Nevertheless, the difference is small, even for a very low choice of $M = 10$, and the discrepancy narrows as M increases. In this exercise, we choose $M = 100$ as the benchmark, balancing both the accuracy of the distribution approximation and the computational speed.

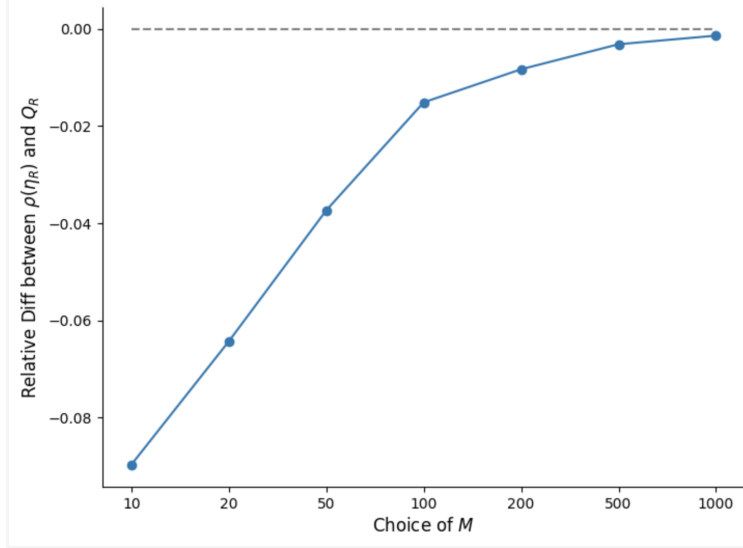


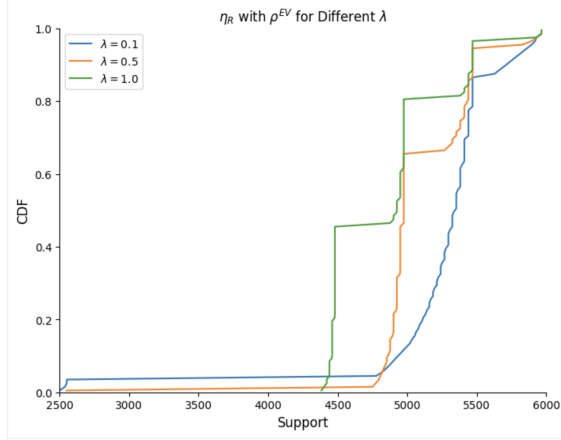
Figure 5: Relative differences between the mean of the rejection-value distribution, $\rho(\eta_R)$, and the VFI benchmark Q_R for $M \in \{10, 20, 50, 100, 200, 1000\}$.

Solutions for different measures ρ . Figure 6 illustrates the solution η_R associated with different choices of ρ , where the agent optimizes not the expected value but an alternative measure of the value distribution function.

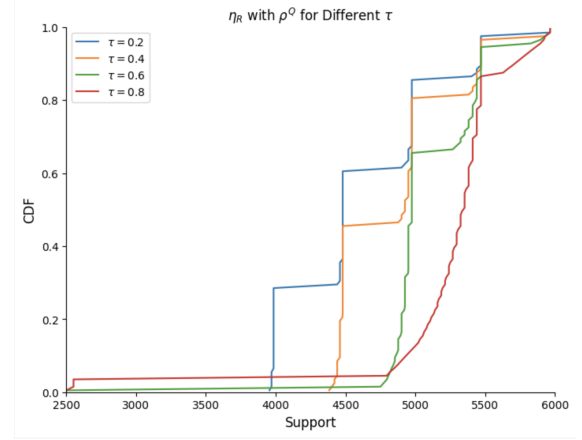
In Figure 6a, we consider the mean–variance measure ρ_{MV}^λ for a range of variance weights, $\lambda \in \{0.1, 0.5, 1.0\}$. The results show that the CDF of η_R exhibits more pronounced steps as λ increases, indicating that the agent is more risk-averse. Another important aspect is that the range of η_R shrinks as λ increases, consistent with the agent’s preference for more certain returns.

In Figure 6b, we examine the quantile measure ρ_Q^τ across different quantile levels, $\tau \in \{0.2, 0.4, 0.6, 0.8\}$. Given that the wage distribution is left-skewed, the η_R associated with higher values of τ is closer to the case where the agent maximizes the expected value. In contrast, the η_R associated with lower values of τ has a higher lower bound, consistent with the agent seeking a higher value at the lower quantile level.

Taken together, the McCall results show that DVFI recovers the VFI solution when the mean measure is used, and that alternative measures—mean–variance and quantile—systematically reshape the rejection-value distribution and, through it, the agent’s accep-



(a) Mean–variance measure ρ_{MV}^λ , $\lambda \in \{0.1, 0.5, 1.0\}$



(b) Quantile measure ρ_Q^τ , $\tau \in \{0.2, 0.4, 0.6, 0.8\}$

Figure 6: CDFs of the rejection-value distribution η_R under alternative measures ρ . Panel (a) varies the variance penalty λ in the mean–variance measure; panel (b) varies the quantile level τ in the quantile measure.

tance behavior. We next examine whether the method carries over to a general-equilibrium setting.

5 Case 2: Aiyagari Model

The Aiyagari model (Aiyagari, 1994) is a foundational framework in macroeconomics that explores the interactions between income inequality, savings behavior, and aggregate economic outcomes. This model extends the standard neoclassical growth model by incorporating heterogeneous agents who face idiosyncratic income shocks and have limited access to financial markets. As a result, it provides valuable insights into the distribution of wealth, the role of precautionary savings, and the effects of incomplete markets on capital accumulation and economic growth. The Aiyagari model is particularly useful for analyzing how individual income risk and borrowing constraints influence aggregate savings and the distribution of wealth in an economy.

5.1 Model Setup

Household

There is a continuum of infinitely lived and ex-ante identical households. In each period, households face a binary productivity shock ϵ_t , which follows an exogenous Markov chain characterized by a transition matrix Π . The households own capital k_t , which they use for

both consumption c_t and investment. Households face a no-borrowing constraint, $k_t \geq 0$, due to incomplete markets. The optimization problem faced by an agent is:

$$\max \mathbb{E} \sum_{t=0}^{\infty} \gamma^t \log(c_t) \quad (15)$$

subject to

$$k_{t+1} + c_t \leq w\epsilon_t + (1+r)k_t \quad (16)$$

$$k_t \geq 0 \quad (17)$$

where w and r denote the wage per efficiency unit of labor and the interest rate, respectively. These two market prices are taken as given from the perspective of households and are determined by aggregation through firms' optimization.

Firms

The goods market is competitive, and firms face constant returns to scale, with production following a Cobb-Douglas form:

$$Y = AK^\alpha L^{1-\alpha} \quad (18)$$

where $A > 0$ and $\alpha \in (0, 1)$ are parameters representing total-factor productivity and the capital share, respectively. $K_t = \int k_t dF(k_t)$ denotes the aggregate capital holdings and L represents the aggregate labor supply, which is exogenously determined by Π . The first-order condition of the firms' maximization problem

$$\max_K \{AK^\alpha L^{1-\alpha} - (r + \delta)K - wL\},$$

where δ denotes the capital depreciation rate, determines the market prices:

$$r = A\alpha \left(\frac{L}{K}\right)^{1-\alpha} - \delta \quad (19)$$

$$w = A(1 - \alpha) \left(\frac{L}{K}\right)^{-\alpha} \quad (20)$$

We can further combine (19) and (20) to express the wage w as a function of the interest rate r :

$$w(r) = A(1 - \alpha) \left(\frac{A\alpha}{r + \delta} \right)^{\alpha/(1-\alpha)} \quad (21)$$

so that we only need to track r as the price in the system. Note that since there are no aggregate shocks in the economy, the distributions of capital holdings and labor productivity are constant over time in equilibrium. Consequently, all the aggregate variables Y , K , and L , as well as the prices r and w , remain constant, allowing us to drop the time subscript.

Equilibrium

We consider a stationary rational expectations equilibrium (SREE) for this problem:

1. Given the supply price r_s (and consequently w), households solve their savings problem (15) subject to (16) and (17), leading to a constant aggregate supply K .
2. The firm's optimization problem (18) leads to the capital demand function (19).
3. The capital market clears and the demand price is consistent with the supply price:
 $r_s = r_d$.

5.2 Computation Strategy

The state space is defined as $\mathcal{S} = [k_{\min}, k_{\max}] \times \{\epsilon_{\text{low}}, \epsilon_{\text{high}}\}$, where the state variable is the pair representing both capital holdings and productivity, $s = (k, \epsilon)$. The action space is $\mathcal{A} = [k_{\min}, k_{\max}]$, with the action being the chosen capital holding for the next period, $a = k'$. The feasibility of k' , given the current state (k, ϵ) , is constrained by the budget constraint of the household (16). We enforce this constraint by defining the reward function $R(s, a)$ such that any state-action pair violating (16) results in a reward value of negative infinity.

The computation of the SREE involves a nested algorithm with an outer and an inner loop.

The outer loop searches for the market-clearing interest rate r^* by bisection. We begin by proposing two supply prices (r_{s1}, r_{s2}) such that the former is too low (leading to excess demand for K , and hence $r_{d1} > r_{s1}$) and the latter is too high ($r_{d2} < r_{s2}$), ensuring that r^* lies within the interval $[r_{s1}, r_{s2}]$. We then evaluate the midpoint of the interval, $r_{s3} = \frac{r_{s1} + r_{s2}}{2}$, and replace r_{s1} or r_{s2} with r_{s3} , depending on whether r_{s3} is too low or too high, thus narrowing the interval. This process is repeated until the interval length falls below a tolerance.

Each evaluation of a candidate price calls the inner loop. Given the supply price r_s , we solve the agent's savings problem using VFI or DVFI to obtain the agent's optimal policy π^* . We then compute the stationary capital distribution $F(K)$ and, consequently, the aggregate

supply K . Finally, we determine the demand price r_d from the firm’s response to K , which tells the outer loop whether the candidate price is too low or too high.

5.3 Calibration and Results

Table 2 lists the structural parameters and hyperparameters chosen for the benchmark case. To discretize the continuous capital dimension $[k_{\min}, k_{\max}]$ of the state space, we choose J grid points $(k_1, \dots, k_J)$, following the polynomial rule (Maliar and Maliar, 2014):

$$k_j = \left(\frac{j}{J}\right)^d (k_{\max} - k_{\min}) + k_{\min} \quad \text{for } j = 1, \dots, J \quad (22)$$

which places more grid points at low capital levels and improves the accuracy of the approximation. The number of grid points J is chosen to balance computational efficiency with the need to sufficiently capture the features of the model.

Parameter	Definition	Value
γ	Discount factor	0.96
α	Capital share	0.33
A	Total-factor productivity	1
L	Aggregate labor supply	1
$(k_{\min}, k_{\max})$	Lower and upper bounds of capital	(0, 10)
$(\epsilon_{\text{low}}, \epsilon_{\text{high}})$	Binary productivity levels	(0.1, 1)
Π	Markov chain characterizing ϵ	$\begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$
d	Degree of the polynomial rule discretization	2
J	Number of grid points in discretization	20

Table 2: Parameters of the Aiyagari model.

Next, we consider the results for different choices of M in the M -quantile parameterization of the distribution, as illustrated in Figure 7. Since visualizing the Q -function and the η -function in this context is challenging, we instead focus on two one-dimensional functions of capital k for illustration.

On the left-hand side of Figure 7, we plot the value functions $V(k, \epsilon)$ obtained after solving for Q and η , for better visualization. We compare the cases of VFI and DVFI with different M values against the capital k on the x-axis, each with two lines corresponding to ϵ_{low} (lower curve) and ϵ_{high} (upper curve). As in Case 1, higher values of M provide a better approximation to the VFI solution, and specifically, when $M = 200$, the difference is almost indistinguishable. On the right-hand side, we plot the stationary distribution function of capital k , characterizing the general equilibrium of the economy obtained from the agents’

optimal policy. The distribution is degenerate, collapsing to a point mass, when M is small. When M is sufficiently high, we approximate the case where there is a mass of hand-to-mouth agents at the lower bound of capital, a key feature of the Aiyagari economy. The distribution closely matches that obtained under VFI when $M = 200$.

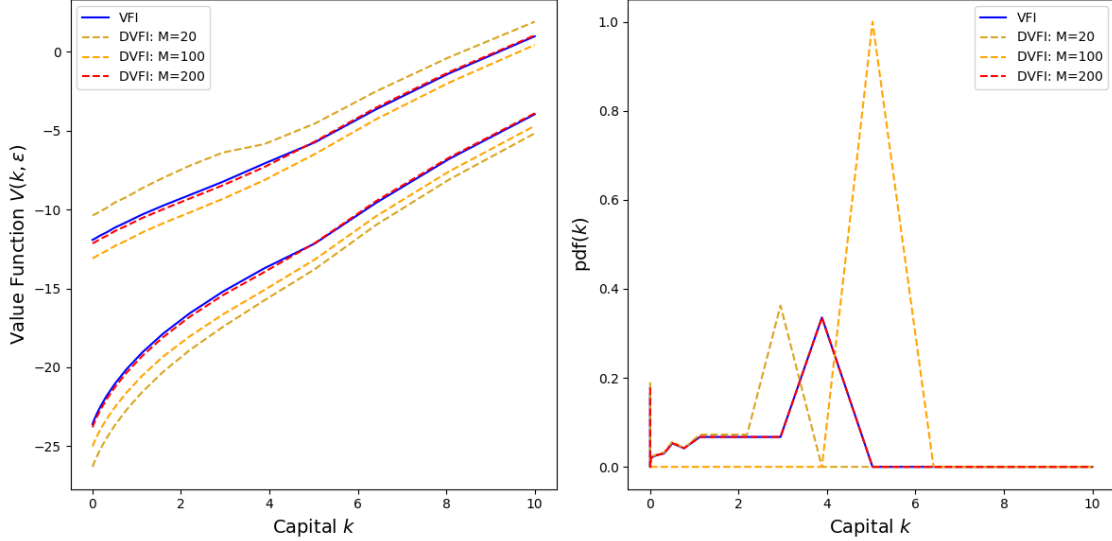


Figure 7: Performance for different choices of M (20, 100, 200). Left: the value functions $V(k, \epsilon)$, each with two lines corresponding to ϵ_{low} (lower curve) and ϵ_{high} (upper curve). Higher values of M provide a better approximation of the correct VFI solution, with $M = 200$ yielding results very close to the exact solution. Right: the stationary asset distribution, $\text{pdf}(k)$. The distribution is more degenerate for smaller M , while for $M = 200$, it closely matches the distribution obtained under VFI.

In sum, the Aiyagari exercise verifies that DVFI reproduces the VFI equilibrium when the mean measure is used, establishing the method’s validity in a general-equilibrium setting. The effect of alternative measures ρ on the stationary wealth distribution is deliberately left for future work; Section 6 outlines this and other extensions.

6 Extensions

Aside from the topics we have already discussed in this paper, there are still many areas for further exploration in future work. Below, we outline three directions:

1. **Parameterized Functional Forms of Distributions:** While this paper focuses on the M -quantile parameterization, representing a distribution with M equally weighted particles at particular quantile levels, other distribution representations might be more advantageous depending on the context and could potentially improve computational efficiency. For example, one possible approach is to adopt a specific functional form for

the distribution and update the parameters either analytically or through incremental methods like gradient descent. However, a challenge is that many such parameterizations are not closed under the distributional Bellman operator. For instance, parameterizing the distribution as normal would result in a mixture of normal distributions after applying the distributional Bellman operator in one iteration. This necessitates a degree of flexibility in the functional form. Neural networks could be a viable option, as there is a growing literature in economics on parameterizing the value function or policy function using neural networks (see, for example, [Azinovic et al., 2022](#); [Maliar et al., 2021](#); [Maliar and Maliar, 2022](#); [Gu et al., 2023](#); [Han and Yang, 2021](#); [Kahou et al., 2021](#)). Parameterizing the value distribution function with a neural network could fall within this domain.

2. **Computational Complexity:** In this paper, we have not extensively discussed the computational complexity and implementation effectiveness of the methods presented, mainly because our focus was on the new economic insights provided by the distributional approach, and the examples we used involve classical models that can be solved quickly, even when extended to the distributional approach. Ideally, the distributional method’s cost is a factor of M times that of the traditional value function approach, since each state-action pair now stores M particles instead of a scalar. However, this depends on the specific algorithm’s implementation, and the cost could change if we consider alternative ways of parameterizing the distribution function, as discussed in the previous point.
3. **Economic Interpretation:** In this paper, we have considered two classical structural models, the McCall search model and the Aiyagari model, both for their established economic insights and because mature routines exist for solving these models using traditional value function approaches, allowing for easy comparison with the distributional approach. Whether the distributional approach can provide enough new insights to justify the additional complexity and slower computation depends on the context. As future extensions, we aim to identify other structural models that might be well-suited to the application of our distributional methods.

7 Conclusion

This paper has introduced Distributional Dynamic Programming, a framework in which agents in structural economic environments evaluate the entire return distribution rather than only its expectation, with preferences encoded by a measure over the distribution function. Building on the *distributional Bellman operator* and the M -quantile parameterization, the proposed *Distributional Value Function Iteration (DVFI)* method provides a practical and theoretically grounded way to solve for the value distribution function. Two applications

illustrate the framework: the McCall search model showed how risk aversion and concern for specific quantiles, encoded as mean–variance and quantile measures, reshape the value distribution and the decision to accept or reject job offers; for the Aiyagari model, we verified that DVFI reproduces the VFI equilibrium when the mean measure is used, establishing the method’s validity in a general-equilibrium setting.

These results carry two practical implications for the modeler. First, the framework generalizes traditional dynamic programming rather than replacing it: when the mean measure is chosen, DVFI recovers the standard solution, so the distributional machinery can be adopted without abandoning existing benchmarks. Second, the approach offers a way to model risk preferences directly through a measure over the return distribution when specifying a suitable utility function is not feasible or straightforward—though this comes at the computational cost of tracking M particles per state-action pair in place of a scalar, so the approach is most attractive where preferences over distributional outcomes are central to the question at hand.

Several directions remain open. A natural next step, deferred in this paper, is to study how alternative measures over the value distribution function affect the stationary wealth distribution in the Aiyagari economy, compared against the expectation-maximizing benchmark. Beyond that, future research could explore applications of the framework to other economic models, investigate alternative parameterizations of the value distribution function—including neural-network representations—and examine the implications of using different measures for policy outcomes. The potential to better capture the complexity of human decision-making under uncertainty makes this a promising direction for ongoing and future work in economics.

References

- Aiyagari, S Rao**, “Uninsured idiosyncratic risk and aggregate saving,” *The Quarterly Journal of Economics*, 1994, *109* (3), 659–684.
- Azinovic, Marlon, Luca Gaegauf, and Simon Scheidegger**, “Deep equilibrium nets,” *International Economic Review*, 2022, *63* (4), 1471–1525.
- Bellemare, Marc G, Will Dabney, and Mark Rowland**, *Distributional reinforcement learning*, MIT Press, 2023.
- , – , and **Rémi Munos**, “A distributional perspective on reinforcement learning,” in “International conference on machine learning” PMLR 2017, pp. 449–458.
- Bellman, Richard**, “Dynamic programming,” *science*, 1966, *153* (3731), 34–37.
- Brock, William A and Leonard J Mirman**, “Optimal economic growth and uncertainty: the discounted case,” *Journal of Economic Theory*, 1972, *4* (3), 479–513.
- Dabney, Will, Georg Ostrovski, David Silver, and Rémi Munos**, “Implicit quantile networks for distributional reinforcement learning,” in “International conference on machine learning” PMLR 2018, pp. 1096–1105.
- , **Mark Rowland, Marc Bellemare, and Rémi Munos**, “Distributional reinforcement learning with quantile regression,” in “Proceedings of the AAAI conference on artificial intelligence,” Vol. 32 2018.
- de Castro, Luciano and Antonio F Galvao**, “Dynamic quantile models of rational behavior,” *Econometrica*, 2019, *87* (6), 1893–1939.
- , – , and **Andre Muchon**, “Numerical Solution of Dynamic Quantile Models,” *Journal of Economic Dynamics and Control*, 2023, *148*, 104617.
- , – , **Charles N Noussair, and Liang Qiao**, “Do people maximize quantiles?,” *Games and economic behavior*, 2022, *132*, 22–40.
- Gu, Zhouzhou, Methieu Lauriere, Sebastian Merkel, and Jonathan Payne**, “Deep learning solutions to master equations for continuous time heterogeneous agent macroeconomic models,” Technical Report 2023.
- Han, Jiequn and Yucheng Yang**, “Deepham: A global solution method for heterogeneous agent models with aggregate shocks,” *arXiv preprint arXiv:2112.14377*, 2021.
- Kahou, Mahdi Ebrahimi, Jesús Fernández-Villaverde, Jesse Perla, and Arnav Sood**, “Exploiting symmetry in high-dimensional dynamic programming,” Technical Report, National Bureau of Economic Research 2021.

- Kantorovich, Leonid V**, “Mathematical methods of organizing and planning production,” *Management science*, 1960, 6 (4), 366–422.
- Koopmans, Tjalling C**, “Stationary ordinal utility and impatience,” *Econometrica: Journal of the Econometric Society*, 1960, pp. 287–309.
- Lyle, Clare, Marc G Bellemare, and Pablo Samuel Castro**, “A comparative analysis of expected and distributional reinforcement learning,” in “Proceedings of the AAAI Conference on Artificial Intelligence,” Vol. 33 2019, pp. 4504–4511.
- Maliar, Lilia and Serguei Maliar**, “Numerical methods for large-scale dynamic economic models,” in “Handbook of computational economics,” Vol. 3, Elsevier, 2014, pp. 325–477.
- **and** –, “Deep learning classification: Modeling discrete labor choice,” *Journal of Economic Dynamics and Control*, 2022, 135, 104295.
- , – , **and Pablo Winant**, “Deep learning for solving dynamic economic models,” *Journal of Monetary Economics*, 2021, 122, 76–101.
- McCall, John Joseph**, “Economics of information and job search,” *The Quarterly Journal of Economics*, 1970, 84 (1), 113–126.
- Samuelson, Paul Anthony**, *Foundations of economic analysis*, Vol. 197, Harvard university press Cambridge, MA, 1983.
- Sargent, Thomas and John Stachurski**, “Quantitative economics,” Technical Report, Citeseer 2014.
- Sutton, Richard S**, “Reinforcement Learning: An Introduction,” 2018.
- Watkins, Christopher J and Peter Dayan**, “Q-learning,” *Machine learning*, 1992, 8, 279–292.